Endocrine system

- The endocrine system is one of the two great control systems of the body , the other being the nervous system
- The endocrine system consists of glands and tissues that secrete hormones.
- They regulate growth, metabolism, reproduction, and behavior.

The endocrine glands are:

- pituitary gland
- thyroid gland
- parathyroid gland
- adrenal gland
- pineal gland
- thymus gland
- The pancreatic islets (islets of langerhans)
- ovaries in the female
- testis in the male

Hormones

- **Definition:**

are chemical messengers that have specific regulatory effects on certain cells or organs.

  - Hormones from the endocrine glands are released directly into the bloodstream, which carries them to all parts of the body (target organ or tissue).
  - They regulate growth, metabolism, reproduction, and behavior.
  - Some hormones affect many tissues, for example, growth hormone, thyroid hormone, and insulin.

Mechanisms of hormone action

- Hormones act by binding to specific receptors either on the target cell surface or within the cell

- Once a hormone binds to a receptor on or in a target cell, it affects cell activities, regulating the manufacture of proteins, changing the permeability of the membrane, or affecting metabolic reactions.

Mechanisms of hormone action

- all hormones produce their effects by altering intracellular protein activity.
- the mechanism by which this occurs depends on the location of the hormone receptor.
- **Receptors are located on**

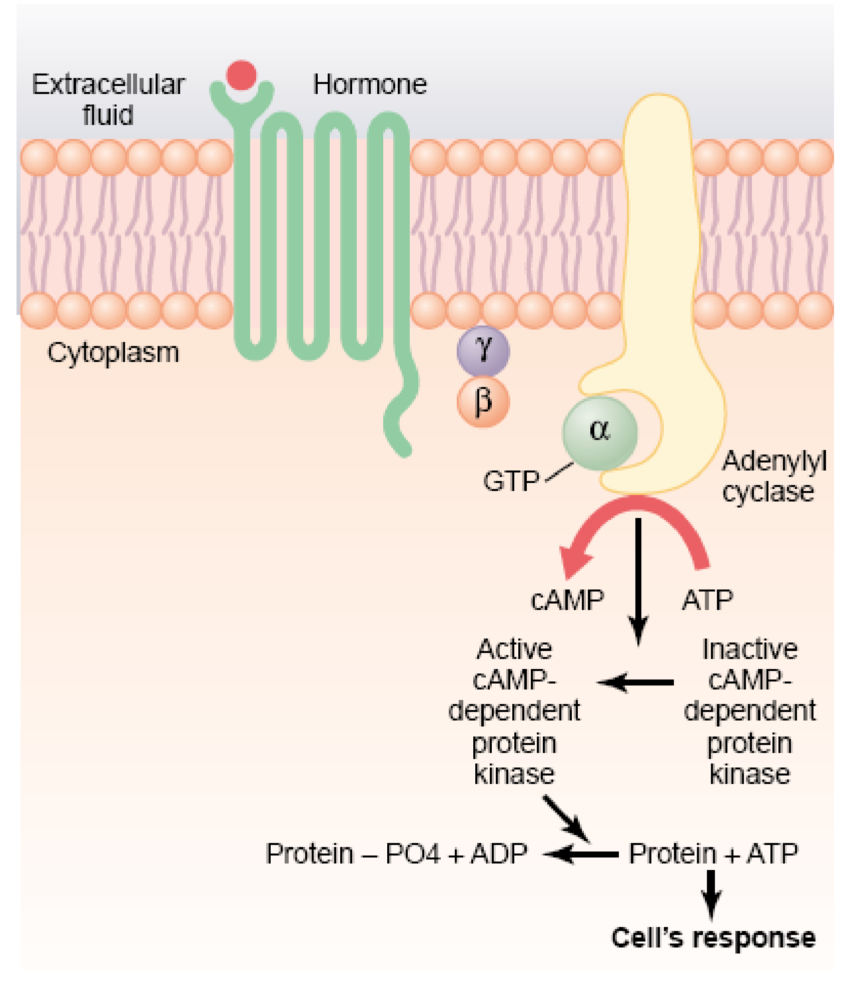
1. **In or on the surface of the cell membrane.** The membrane receptors are specific mostly for the protein, peptide, and catecholamine hormones.
2. **In the cell cytoplasm.** The primary receptors for the different steroid hormones are found mainly in the cytoplasm.
3. **In the cell nucleus.** The receptors for the thyroid hormones are found in the nucleus and are believed to be located in direct association with one or more of the chromosomes.

- hormones carry out their effects by the following mechanisms:

1) **Adenylyl Cyclase–cAMP Second Messenger System**

- Hormones exert intracellular actions by stimulate formation of the second messenger inside the cell membrane like
   - A. Cyclic adenosine monophosphate (cAMP)
   - B. calcium ions and associated calmodulin ,
   - C. products of membrane phospholipid breakdown.
- **Second Messenger** causes subsequent intracellular effects of the hormone.

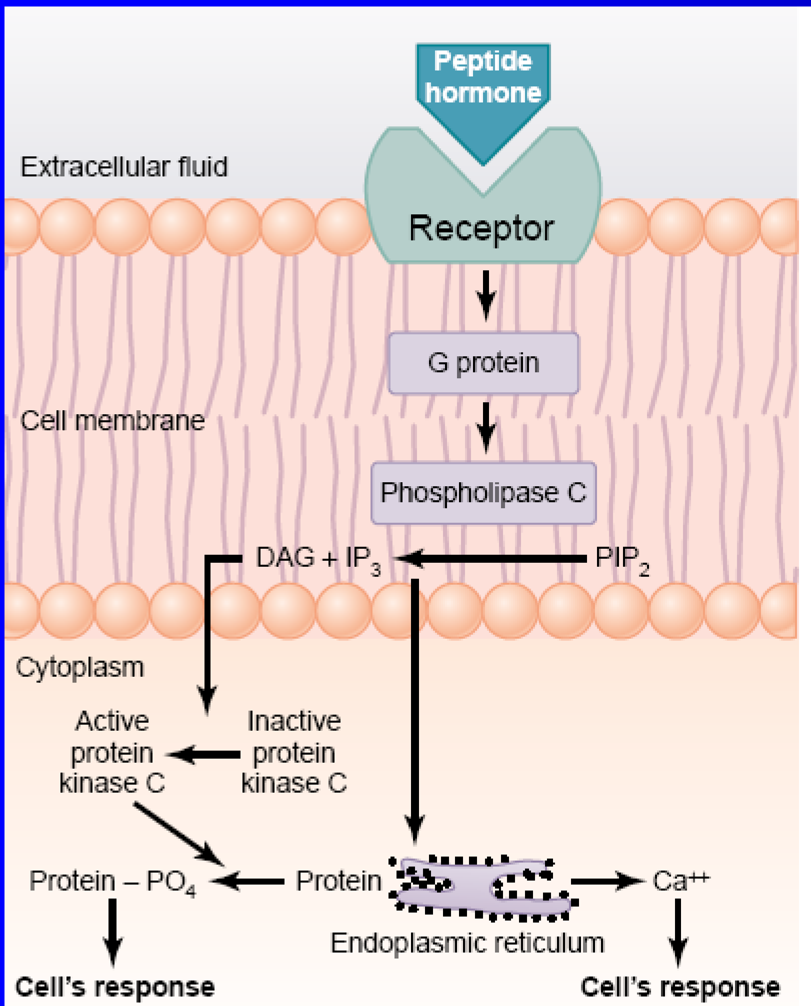
Cyclic adenosine monophosphate (cAMP) mechanism by which many hormones exert their control of cell function. ADP (adenosine diphosphate); ATP (adenosine triphosphate).



Some Hormones That Use the Adenylyl Cyclase–cAMP Second Messenger System

Adrenocorticotropic hormone (ACTH)
Angiotensin II (epithelial cells)
Calcitonin
Catecholamines (β receptors)
Corticotropin-releasing hormone (CRH)
Follicle-stimulating hormone (FSH)
Glucagon
Human chorionic gonadotropin (HCG)
Luteinizing hormone (LH)
Parathyroid hormone (PTH)
Secretin
Somatostatin
Thyroid-stimulating hormone (TSH)
Vasopressin (V_2 receptor, epithelial cells)

2) The Cell Membrane Phospholipid Second Messenger System



Some Hormones That Use the Phospholipase C Second Messenger System
Angiotensin II (vascular smooth muscle)
Catecholamines (α receptors)
Gonadotropin-releasing hormone (GnRH)
Growth hormone–releasing hormone (GHRH)
Oxytocin
Thyroid-releasing hormone (TRH)
Vasopressin (V_1 receptor, vascular smooth muscle)

The cell membrane phospholipid second messenger system by which some hormones exert their control of cell function. DAG, diacylglycerol; IP3(inositol triphosphate); PIP2(phosphatidylinositol biphosphate).

Hormones – chemical structure

There are three general classes of hormones:

1. **Proteins and polypeptides**

– the anterior and posterior pituitary gland hormones, (insulin, glucagon and parathyroidal hormone).

- 3 or more amino acids - Water soluble

2. **Steroids**

–cortisol, aldosterone), estrogen, progesterone, testosterone

- synthesized from cholesterol
- Lipid soluble

3. **Amino hormones**

- Derivates of amino acid tyrosine

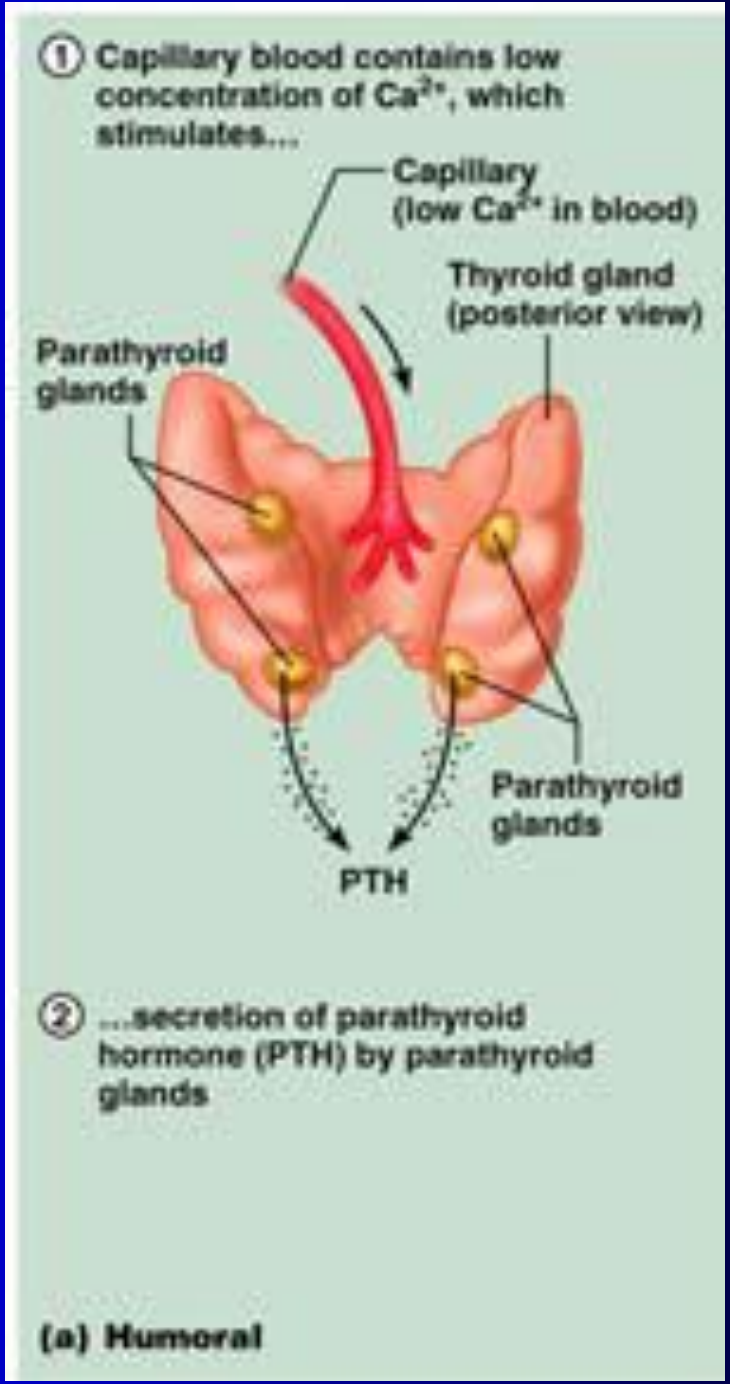
– thyroxine, triiodothyronine, epinephrine and norepinephrine)

- Hormones are synthesized and released in response to:
    - Humoral stimuli
    - Neural stimuli
    - Hormonal stimuli

Humoral Stimuli

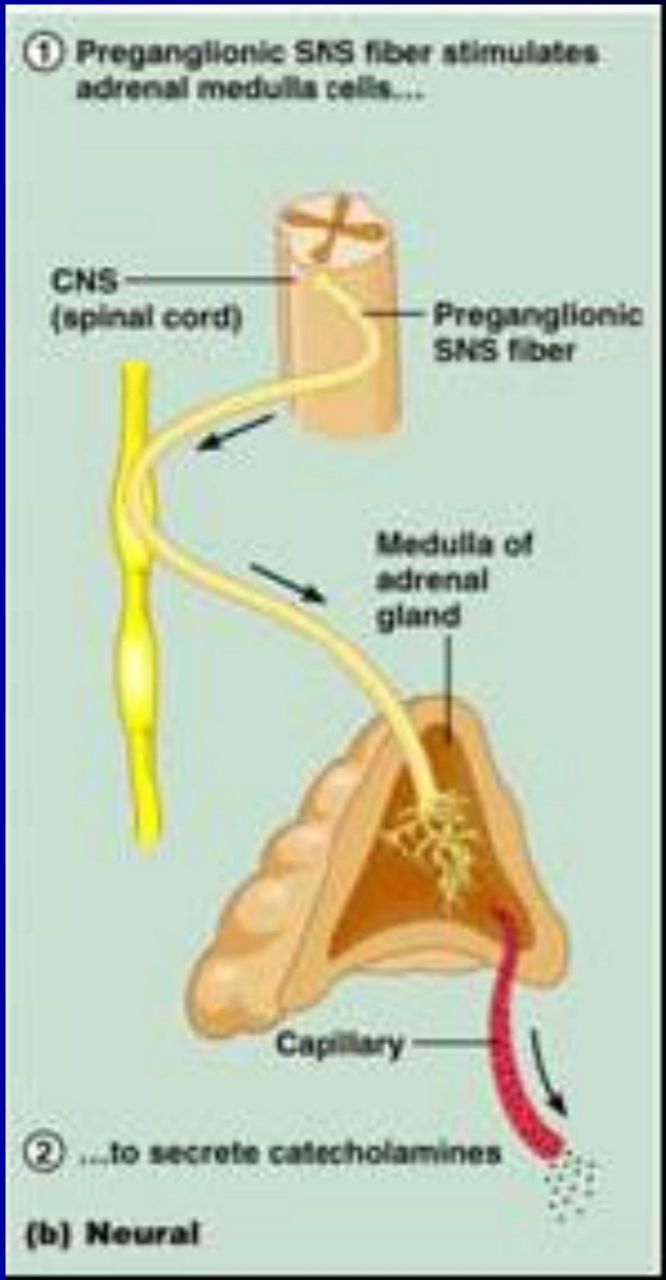
- Humoral stimuli – secretion of hormones in direct response to changing blood levels of ions and nutrients
- Example: concentration of calcium ions in the blood
  - Declining blood Ca^{+2} concentration stimulates the parathyroid glands to secrete PTH (parathyroid hormone)
  - PTH causes Ca^{+2} concentrations to rise and the stimulus is removed

Humoral stmuli



Neural Stimuli

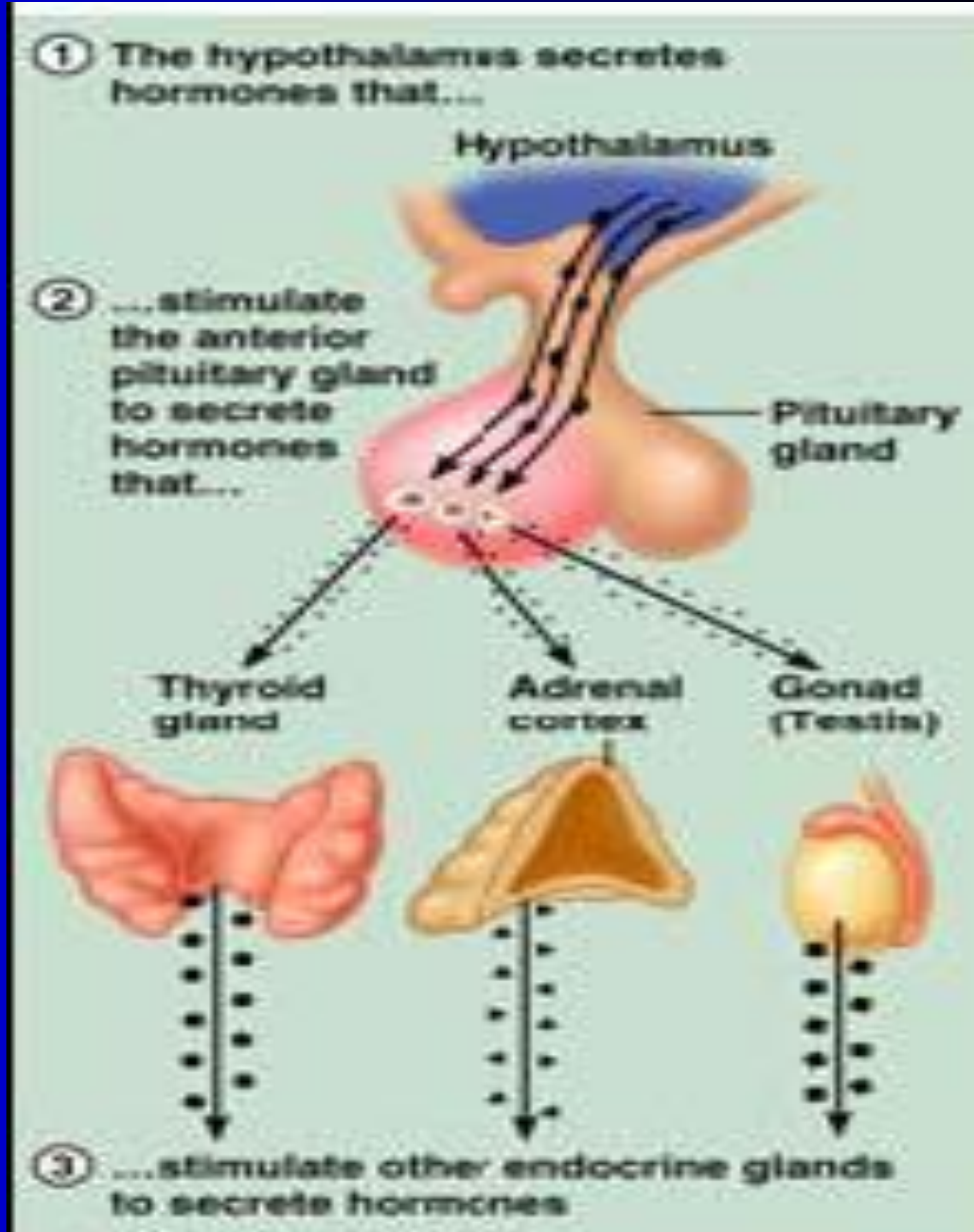
- Neural stimuli – nerve fibers stimulate hormone release
    - Preganglionic sympathetic nervous system (SNS) fibers stimulate the adrenal medulla to secrete catecholamines



Hormonal Stimuli

- Hormonal stimuli – release of hormones in response to hormones produced by other endocrine organs
  - The hypothalamic hormones stimulate the anterior pituitary
  - In turn, pituitary hormones stimulate targets to secrete still more hormones

Hormonal stimuli



(c) Hormonal

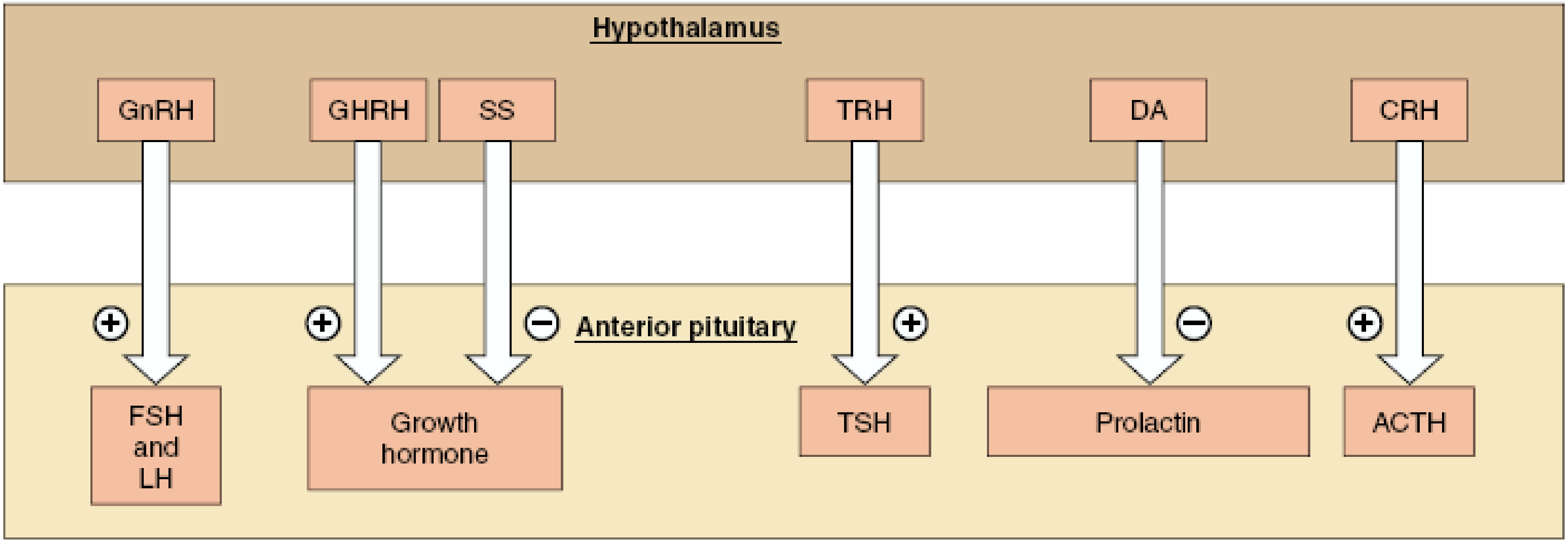
Pituitary gland (Hypophysis)

- Regulating the activity of most the other endocrine glands(so called master gland)
- Consists of 3 distinct parts(anterior, posterior and intermediate lobes)
- **Neurohypophysis (posterior lobe):** neural tissue and the infundibulum
  - Receives, stores, and releases hormones from the hypothalamus
- **Adenohypophysis (anterior lobe):** made up of glandular tissue
  - Synthesizes and secretes a number of hormones

Control of anterior pituitary secretion

1. **The hypothalamic control:**
   - Through the hypothalamic releasing and inhibitory factors (hormones) secreted by hypothalamus and carried to the anterior pituitary through the hypothalamic hypophyseal portal circulation
2. **Feed back control:**

   blood levels of adrenocortical hormones, thyroid hormones and gonad steroid influence the pituitary secretion of ACTH, TSH and gonadotropins respectively. The influence in mainly inhibitory in nature (negative feedback mechanism)



Major known hypophysiotropic hormones	Major effect on anterior pituitary
Corticotropin releasing hormone (CRH)	Stimulates secretion of ACTH
Thyrotropin releasing hormone (TRH)*	Stimulates secretion of TSH
Growth hormone releasing hormone (GHRH)	Stimulates secretion of GH
Somatostatin (SS, also known as growth hormone release inhibiting hormone, GIH)	Inhibits secretion of GH
Gonadotropin releasing hormone (GnRH)	Stimulates secretion of LH and FSH
Dopamine (DA, also known as prolactin-inhibiting hormone, PIH)‡	Inhibits secretion of prolactin

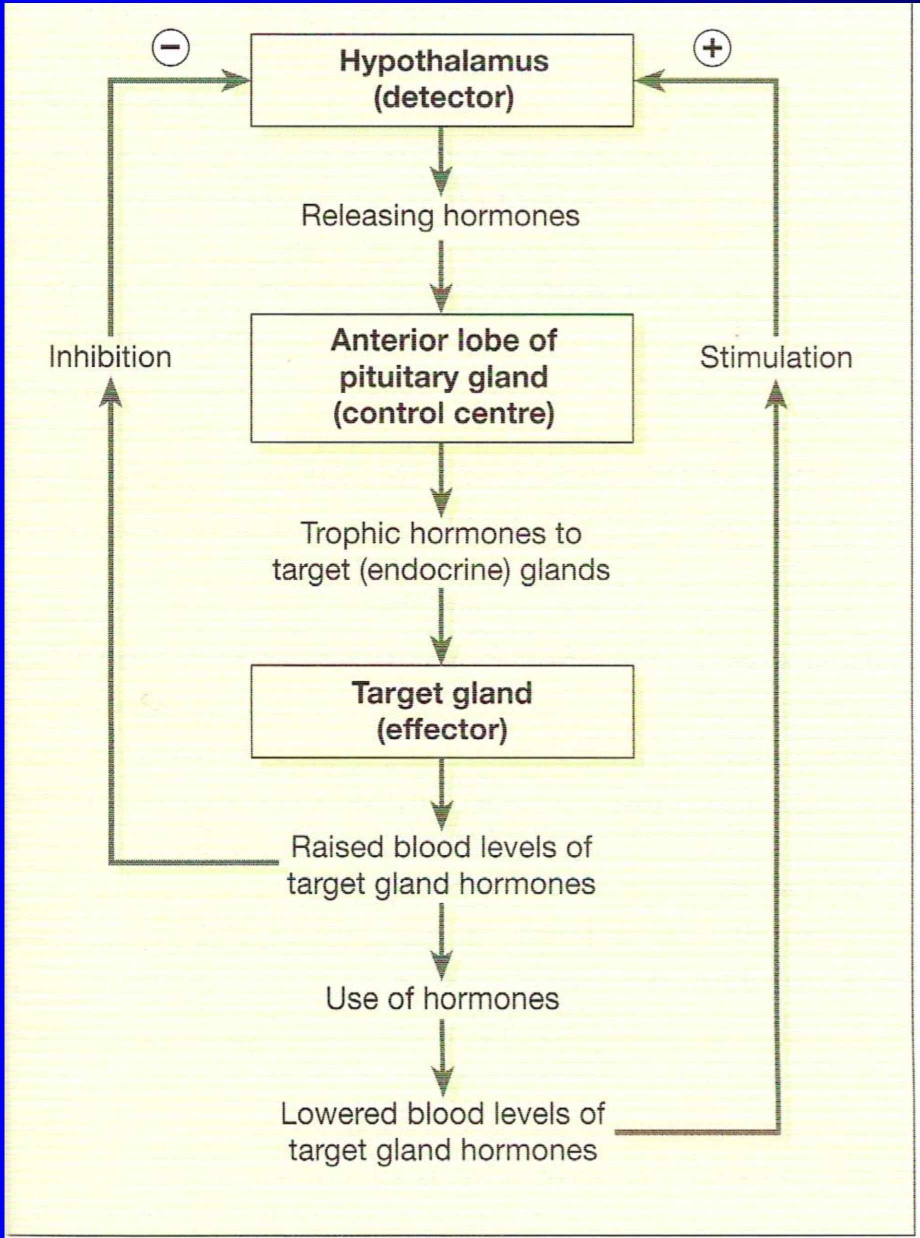
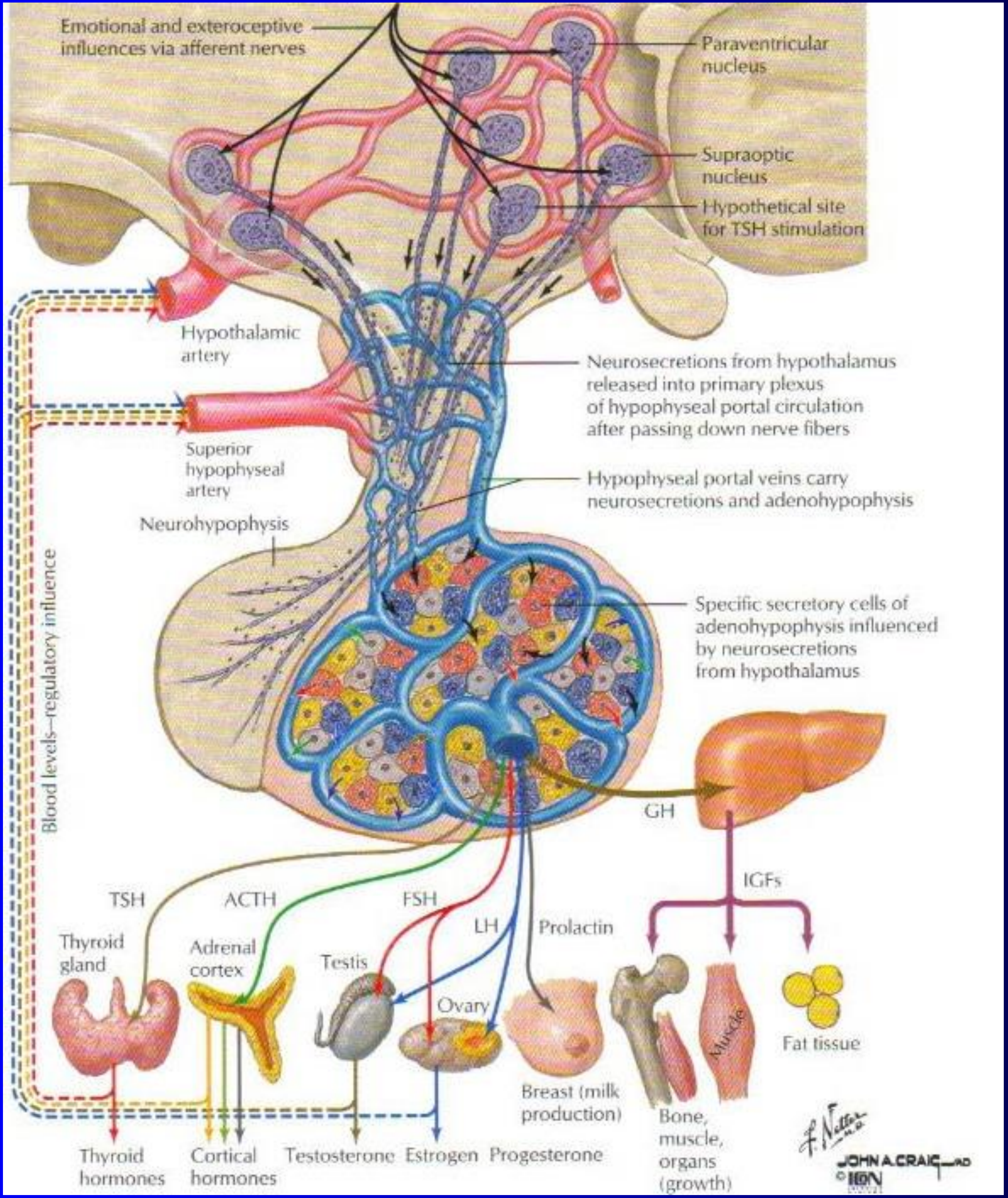
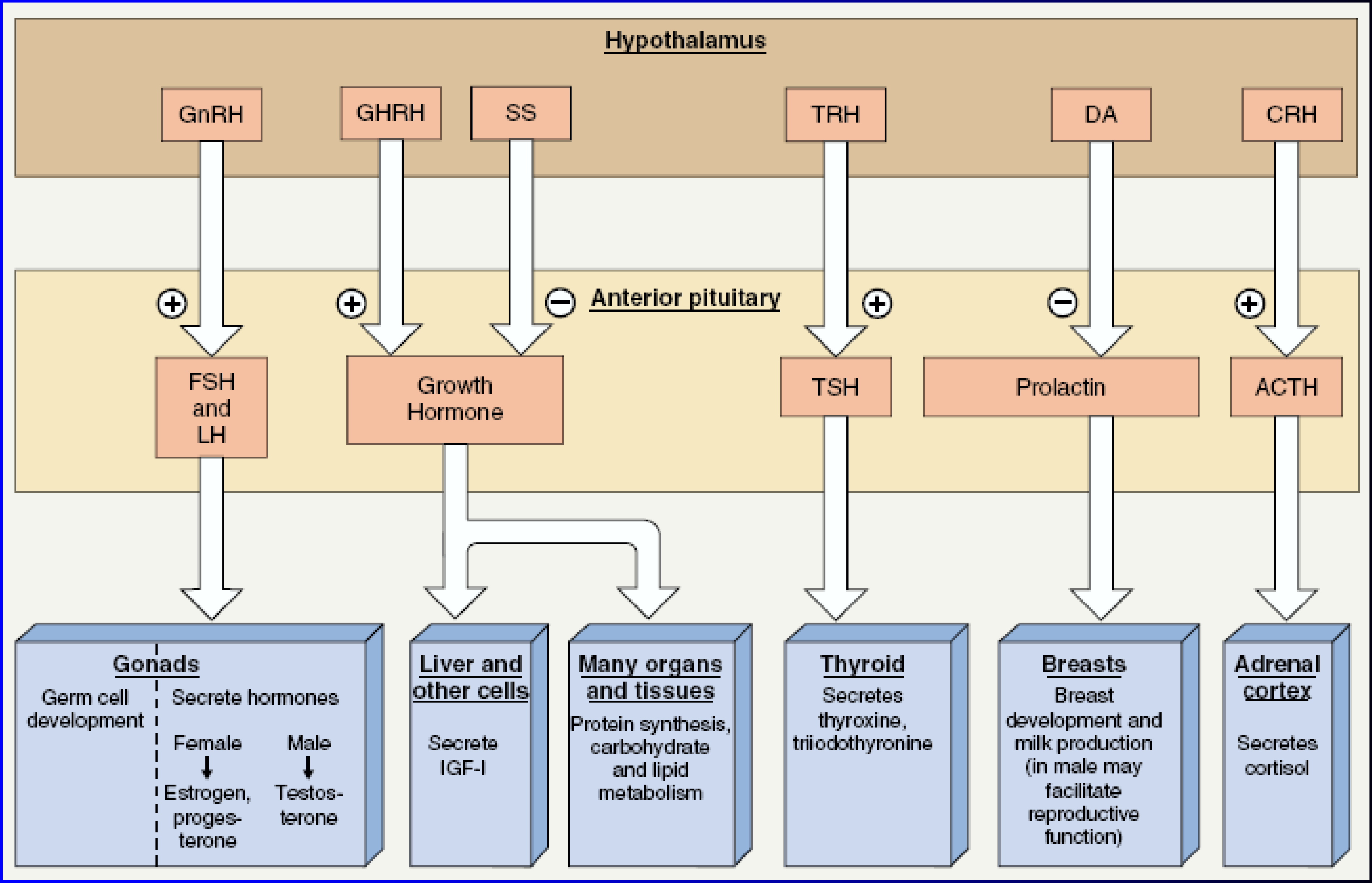


Figure 9.4 Negative feedback regulation of secretion of hormones by the anterior lobe of the pituitary gland.


Emotional and exteroceptive influences via afferent nerves
Paraventricular nucleus
Supraoptic nucleus
Hypothetical site for TSH stimulation
Hypothalamic artery
Neurosecretions from hypothalamus released into primary plexus of hypophyseal portal circulation after passing down nerve fibers
Superior hypophyseal artery
Hypophyseal portal veins carry neurosecretions and adenohypophysis
Neurohypophysis
Specific secretory cells of adenohypophysis influenced by neurosecretions from hypothalamus
Blood levels—regulatory influence
GH
IGFs
TSH
ACTH
FSH
LH
Prolactin
Thyroid gland
Adrenal cortex
Testis
Ovary
Breast (milk production)
Muscle
Fat tissue
Bone, muscle, organs (growth)
Thyroid hormones
Cortical hormones
Testosterone
Estrogen
Progesterone
JOHN A.CRAIG



Hypothalamus
GnRH
GHRH
SS
TRH
DA
CRH
Anterior pituitary
FSH and LH
Growth Hormone
TSH
Prolactin
ACTH
Gonads
Germ cell development
Secrete hormones
Female
Estrogen, proges-terone
Male
Testos-terone
Liver and other cells
Secrete IGF-I
Many organs and tissues
Protein synthesis, carbohydrate and lipid metabolism
Thyroid
Secretes thyroxine, triiodothyronine
Breasts
Breast development and milk production (in male may facilitate reproductive function)
Adrenal cortex
Secretes cortisol

Hormone secreted by anterior lobe of the pituitary gland

1. Growth hormone(GH)

- **Function**
  - Stimulate growth and division of most cell body cells (bones &skeletal muscle)
  - Regulate aspects of metabolism in liver, intestines and pancreas
  - Stimulate protein synthesis, promote breakdown of fats and increase blood glucose levels
- **Stimulated by:**
  - GHRH
  - Hypoglycemia, exercise ,increase level of amino acid in blood
  - Low levels of fatty acid
  - estrogens

- **Inhibited by:**
  - Negative feedback mechanism when the blood level rises and when GHIH released from hypothalamus
  - Hyperglycemia and obesity

- **Effects of excess GH**
- **Gigantism:** this occurs when there is excess GH during childhood before ossification of bones is complete
- **Acromegaly:** this occurs when there is excess GH after ossification is complete (an enlarged tongue and excessively large hands and feet)
- **Pituitary dwarfism:** this caused by severe deficiency of GH in childhood

- **Thyroid stimulating hormone (TSH)**
  - During the night the release of hormone is highest and lowest in early evening
- **Function:**
  - It Stimulates growth and activity of the thyroid gland
- **Stimulated by:**
  - TRH and indirectly by pregnancy and cold temperature
- **Inhibited by:**
  - Feedback inhibition when the blood level of thyroid is high (T_4 and T_3), secretion of TSH is reduced

3. **Adrenocorticotrophic hormone (ACTH)**
   - Increases the concentration of cholesterol and steroid hormones within the adrenal cortex and the output of steriod hormones especially cortisol

- **Stimulated by:**
  - CRH(corticotrophin);stimuli that increase CRH include fever and hypoglycemia

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- ACTH levels are highest at about 8 a.m and fall to their lowest about midnight

- **Inhibited by:**
  - feedback inhibition exerted by glucocorticoids

4. Prolactin

- **Function:**
  - stimulate lactation (milk production)
  - Has a direct effect on the breasts after childbirth
  - The resultant high blood level is a factor in reducing a conception rate during lactation

- **Stimulated by:**
  - PRH released from hypothalamus
  - Suckling stimulates prolactin secretion and lactation
- **Inhibited by:**
  - PIH and an increased blood level of PR
  - **Note:** PR is raised during any period of sleep, night or day
  - PR together with oestrogens, corticosteroids, insulin and thyroxin is involved in initiating and maintaining lactation

5. **Gonadotrophins (sex hormones)(FSH and LH)**
   - **They are secreted by anterior lobe after puberty**
- **Stimulated by:** LHRH(*luteinizing hormone releasing hormone*) **or** GnRH (*gonadotrophin releasing hormone)*
- **Function:**
  - **In both sexes. FSH** stimulates production of gametes (ova or spermatozoa)
  - **In females.** LH and FSH are involved in secretion of hormones oestrogen and progesterone during the menstrual cycle

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- As the levels of oestrogen and progesterone rise, secretion of LH and FSH suppresed
- **In males.** LH stimulates the interstitial cells of the testis to secrete the hormone testosterone

Posterior pituitary Hormones

- Mad by hypothalamic neurons and stored in posterior lobe
- These hormones act directly on non-endocrine tissue

1. **Oxytocin**

- **Stimulated by:** impulses from hypothalamic neurons in response to cervical/ uterine stretching and suckling of infant at breast (this an example *of positive feedback mechanism* which stopped after the baby is delivered)

- **Inhibited by:** lack of appropriate neural stimuli
- **Function:**
- **Uterine:**

- Stimulates uterine contractions; initiates labor during parturition
- **Breast:**
- Stimulates contraction of the myoepthelial cells around the glandular cells and ducts of the lactating breast to contract, ejecting milk

Regulation of secretion of oxytocin through positive feedback mechanism

In labour, uterine contractions force the baby,s head into the cervix

+

Stretch receptors in utrine cervix (detector)

Nerve impulses

Hypothalamus and posterior pituitary (control center)

Release of oxytocin

+

Uterine smooth muscle (effector)

Stronger contractions force the baby,s head further into the cervix

Great stimulation

+

Inhibition occurs after delivery when uterine contractions no longer dilate (stretch) the cervix

2. Antidiuretic hormone (ADH) or vasopressin

- **Function:**
  - Stimulates kidney tubule cells to reabsorb water especially distal convoluted and collecting tubules of the nephrons
  - Reduce urine output
- **Stimulated by:**
  - impulses from hypothalamic neurons in response to osmolality of blood or decreased blood volume also stimulated by low blood pressure.
- **Inhibited by:**
  - Low of the osmotic pressure of the blood (for example after a large fluid intake)